
PETIMOT: A Novel Framework for Inferring Protein Motions from Sparse Data Using SE(3)-Equivariant Graph Neural Networks

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Abstract

1 Proteins move and deform to ensure their biological functions. Despite significant
2 progress in protein structure prediction, approximating conformational ensembles at
3 physiological conditions remains a fundamental open problem. This paper presents
4 a novel perspective on the problem by directly targeting continuous compact rep-
5 resentations of protein motions inferred from sparse experimental observations.
6 We develop a task-specific loss function enforcing data symmetries, including
7 scaling and permutation operations. Our method PETIMOT (Protein sEquence and
8 sTructure-based Inference of MOTions) leverages transfer learning from pre-trained
9 protein language models through an SE(3)-equivariant graph neural network. When
10 trained and evaluated on the Protein Data Bank, PETIMOT shows superior perfor-
11 mance in time and accuracy, capturing protein dynamics, particularly large/slow
12 conformational changes, compared to state-of-the-art flow-matching approaches
13 and traditional physics-based models. *Our code and protocols are available at*
14 *<https://anonymous.4open.science/r/PETIMOT-4ED4/>.*

15 1 Introduction

16 Proteins orchestrate biological processes in living organisms by interacting with their environment
17 and adapting their three-dimensional (3D) structures to engage with cellular partners, including
18 other proteins, nucleic acids, small-molecule ligands, and co-factors. In recent years, spectacular
19 advances in high-throughput deep learning (DL) technologies have provided access to reliable
20 predictions of protein 3D structures at the scale of entire proteomes [1]. These breakthroughs have
21 also highlighted the complexities of protein conformational heterogeneity. State-of-the-art predictors
22 struggle to model alternative conformations, fold switches, large-amplitude conformational changes,
23 and solution ensembles [2].

24 The success of AlphaFold2 [3] has stimulated machine-learning approaches focused on inference-
25 time interventions in the model to generate structural diversity. They include enabling or increasing
26 dropout [4, 5], or manipulating the evolutionary information given as input to the model [6–9].
27 Despite promising results on specific families, several studies have emphasised the difficulties in
28 rationalising the effectiveness of these modifications and interpreting them [10, 11]. Moreover,
29 these cannot be transferred to protein language model-based predictors that do not rely on multiple
30 sequence alignments. Researchers have also actively engaged in the development of deep-learning
31 frameworks based on diffusion, or the more general flow matching, to generate conformational
32 ensembles [12]. While family-specific models proved useful in exploring native-like conformational
33 landscapes, models trained across protein families still fail to approximate solution ensembles [13].

This work presents a new glance at the protein conformational diversity problem. Instead of learning and sampling from multi-dimensional empirical distributions, we propose to learn eigenspaces (the structure) of the positional covariance matrices in collections of experimental 3D structures and generalize these over different homology levels. Our motivation is that the diversity present within different 3D structures of the same protein or close homologs is a good proxy for the conformational heterogeneity of proteins in solution [14] and can generally be (almost fully) explained by a small set of linear vectors, also referred to as modes [15, 16]. Although linear spaces may not be well-suited for capturing highly complex non-linear motions, such as loop deformations, they offer multiple advantages. These include faster learning due to the reduced complexity of the model, improved explainability as the components directly correspond to interpretable data dimensions, faster inference, and the straightforward combination or integration of multiple data dimensions.

To summarize, our main contributions are:

- We provide a novel formulation of the protein conformational diversity problem.
- We present a novel benchmark representative of the Protein Data Bank structural diversity and compiled with a robust pipeline [15], along with data- and task-specific metrics.
- We develop a SE(3)-equivariant Graph Neural Network architecture equipped with a novel symmetry-aware loss function for comparing linear subspaces, with invariance to permutation and scaling. Our model, PETIMOT, leverages embeddings from pre-trained protein language models (pLMs), building on prior proof-of-concept work demonstrating that they encode information about functional protein motions [17].
- PETIMOT is trained on sparse experimental data without any use of simulation data, in contrast with Timewarp for instance [18]. Moreover, our model does not require physics-based guidance or feedback, unlike [12] for instance.
- Our results demonstrate the capability of PETIMOT to generalise across protein families (contrary to variational autoencoder-based approaches) and to compare favorably in running time and accuracy to AlphaFlow, ESMFlow, and the Normal Mode Analysis.

2 Related Works

Protein structure prediction. AlphaFold2 was the first end-to-end deep neural network to achieve near-experimental accuracy in predicting protein 3D structures, even for challenging cases with low sequence similarity to proteins with resolved structures [3]. It extracts information from an input multiple sequence alignment (MSA) and outputs all-atom 3D coordinates. Later works have shown that substituting the input alignment by embeddings from a protein language model can yield comparable performance [19–22].

Generating conformational ensembles. Beyond the single-structure frontier, several studies have underscored the limitations and potential of protein structure predictors (PSP) for generating alternative conformations [23, 24, 11, 2]. Approaches focused on re-purposing AlphaFold2 include dropout-based massive sampling [4, 5], guiding the predictions with state-annotated templates [25, 26], and inputting shallow, masked, corrupted, subsampled or clustered alignments [6–9]. Despite promising results, these approaches remain computationally expensive and their generalisability, interpretability, and controllability remain unclear [11, 2]. More recent works have aimed at overcoming these limitations by directly optimising PSP learnt embeddings under low-dimensional ensemble constraints [27].

Another line of research has consisted in fine-tuning or re-training AlphaFold2 and other single-state PSP under diffusion or flow matching frameworks [28, 13, 29]. For instance, the AlphaFlow/ESMFlow method progressively denoises samples drawn from a harmonic prior under flow field controlled by AlphaFold or ESMFold [28]. It compares favourably with MSA subsampling or clustering baselines, with a substantially superior precision-diversity Pareto frontier. More generally, diffusion- and flow matching-based models allow for efficiently generating diverse conformations conditioned on the presence of ligands or cellular partners [30, 31, 12, 32, 33]. Despite their strengths, these techniques are prone to hallucination.

Parallel related works have sought to directly learn generative models of equilibrium Boltzmann distributions using normalising flows [34, 18], or machine-learning force fields based on equivariant

graph neural network (GNN) representations [35], to enhance or replace molecular dynamics (MD) simulations.

Protein conformational heterogeneity manifold learning. Unsupervised, physics-based Normal Mode Analysis (NMA) has long been effective for inferring functional modes of deformation by leveraging the topology of a single protein 3D structure [36–38]. While appealing for its computational efficiency, the accuracy of NMA strongly depends on the initial topology [39], limiting its ability to model extensive secondary structure rearrangements. Recent efforts have sought to address these limitations by directly learning continuous, compact representations of protein motions from sparse experimental 3D structures. These approaches employ dimensionality reduction techniques, from classical manifold learning methods [15] to neural network architectures like variational auto-encoders [40]. By projecting motions onto a learned low-dimensional manifold, these methods enable reconstruction of accurate, physico-chemically realistic conformations, both within the interpolation regime and near the convex hull of the training data [15]. Additionally, they assist in identifying collective variables from molecular dynamics (MD) simulations, supporting importance-sampling strategies [41–45]. Despite these advances, such approaches are currently constrained to family-specific models.

E(3)-equivariant graph neural networks. Graph Neural Networks (GNN) have been extensively used to represent protein 3D structures. They are robust to transformations of the Euclidean group, namely rotations, reflections, and translations, as well as to permutations. In their simplest formulation, each node represents an atom and any pair of atoms are connected by an edge if their distance is smaller than a cutoff or among the smallest k interatomic distances. Many works have proposed to enrich this graph representation with SE(3)-equivariant features informing the model about interatomic directions and orientations [46–49, 31, 50]. For instance, VisNet captures the full local geometric information, including bonds, angles, as well as dihedral torsion and improper angles with node-wise high-order geometric tensors [50]. Moreover, to go beyond local 3D neighbourhoods while maintaining sub-quadratic complexity, Chroma adds in randomly sampled long-range connections [31].

3 Methods

3.1 Data representation

To generate training data, we exploit experimental protein single chain structures available in the Protein Data Bank (PDB) [51]. We first clustered these chains based on their sequence similarity. Then, within each cluster, we aligned the protein sequences and used the resulting mapping for superimposing the 3D coordinates [15]. It may happen that some residues in the multiple sequence alignment do not have resolved 3D coordinates in all conformations. To account for this uncertainty, we assigned a confidence score w_i to each residue i computed as the proportion of conformations including this residue. The 3D superimposition puts the conformations’ centers of mass to zero and then aims at determining the optimal least-squares rotation minimizing the Root Mean Square Deviation (RMSD) between any conformation and a reference conformation, while accounting for the confidence scores [52, 53],

$$E = \frac{1}{\sum_i w_i} \sum_i w_i (\vec{r}_{ij} - \vec{r}_{i0})^2, \quad (1)$$

where $\vec{r}_{ij} \in \mathbb{R}^3$ is the i th centred coordinate of the j th conformation and $\vec{r}_{i0} \in \mathbb{R}^3$ is the i th centred coordinate of the reference conformation. Next, we defined our ground-truth targets as eigenspaces of the coverage-weighted C α -atom positional covariance matrix,

$$C = \frac{1}{m-1} R^c W (R^c)^T = \frac{1}{m-1} (R - R^0) W (R - R^0)^T, \quad (2)$$

where R is the $3N \times m$ positional matrix with N the number of residues and m is the number of conformations, R^0 contains the coordinates of the reference conformation, and W is the $3N \times 3N$ diagonal coverage matrix. The covariance matrix is a $3N \times 3N$ square matrix, symmetric and real. We decompose C as $C = Y D Y^T$, where Y is a $3N \times 3N$ matrix with each column defining a coverage-weighted eigenvector or a principal component that we interpret as a *linear motion*. D is a

133 diagonal matrix containing the eigenvalues. The latter highly depend on the sampling biases in the
 134 PDB and thus we do not aim at predicting them.

135 3.2 Problem formulation

136 For a protein of length N , let Y be $3N \times K$ *orthogonal* ground-truth deformations,

$$Y^T Y = I_K. \quad (3)$$

137 Our goal is to find coverage-weighted vectors $X \in \mathbb{R}^{3N \times L}$ whose components l *approximate* some
 138 components k of the ground truth Y :

$$W^{\frac{1}{2}} \tilde{\mathbf{x}}_l \approx \mathbf{y}_k. \quad (4)$$

139 Below, we provide three alternative formulations for this problem.

140 3.3 Geometric Loss

141 **The least-square formulation.** PETIMOT’s loss function serves two key purposes: it enables
 142 effective training of the network to predict subspaces representing multiple distinct modes of deforma-
 143 tions – *i.e.*, with low overlap between the subspace’s individual linear vectors, while preventing
 144 convergence to a single dominant mode. For each protein of length N with a coverage W , we
 145 compare ground-truth directions Y with predicted motion directions X by computing a weighted
 146 pairwise *least-square difference* $\mathcal{L}_{kl}$ for each pair of a k direction in the ground truth and an l direction
 147 in the prediction,

$$\mathcal{L}_{kl} = \frac{1}{N} \sum_{i=1}^N \|\vec{y}_{ik} - w_i^{1/2} c_{kl} \vec{x}_{il}\|^2 = \frac{1}{N} \mathbf{y}_k^T \mathbf{y}_k - \frac{1}{N} \frac{(\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l)^2}{\mathbf{x}_l^T W \mathbf{x}_l}, \quad (5)$$

148 where we scaled the ground-truth tensors such that $Y^T Y = N I_K$ and we used the fact that the
 149 optimal scaling coefficients c_{kl} between the k -th ground truth vector and the l -th prediction are given
 150 by

$$c_{kl} = \frac{\sum_{i=1}^N w_i^{\frac{1}{2}} \mathbf{y}_{ik}^T \mathbf{x}_{il}}{\sum_{i=1}^N w_i \mathbf{x}_{il}^T \mathbf{x}_{il}} = \frac{\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l}{\mathbf{x}_l^T W \mathbf{x}_l}. \quad (6)$$

151 This invariance to global scaling is motivated by the fact that we aim at capturing the relative
 152 magnitudes and directions of the motion patterns rather than their sign or absolute amplitudes.

153 **Linear assignment problem.** We then formulate an *optimal linear assignment problem* to find the
 154 minimum-cost matching between the ground-truth and the predicted directions. Specifically, we aim
 155 to solve the following assignment problem for the least-square (LS) costs,

$$\text{LS Loss} = \min_{\pi \in S_J} \sum_{k=1}^{\min(K,L)} \mathcal{L}_{k, \pi(k)} \quad (7)$$

subject to:

$$\pi : \{1, \dots, \min(K, L)\} \rightarrow \{1, \dots, L\}, \quad \pi(k) \neq \pi(k') \text{ for } k \neq k',$$

156 where K and L are the number of ground-truth and predicted directions respectively, and $\pi(k)$
 157 represents the index of the predicted direction assigned to the k -th ground truth direction. This
 158 formulation ensures an optimal one-to-one matching, while accommodating cases where the number
 159 of predicted and ground-truth directions differs. We backpropagate the loss only through the optimally
 160 matched pairs, using *scipy* `linear_sum_assignment`. We have also tested a smooth version of the loss
 161 above with continuous gradients, but it did not improve the performance.

162 **The subspace coverage formulation.** We propose another formulation of the problem in terms of
 163 the subspace coverage metrics [54–56]. Specifically, we sum up *squared sinus* (SS) dissimilarities

164 between ground-truth and predicted directions (formally computed as one minus squared cosine
165 similarity),

$$\text{SS Loss} = 1 - \frac{1}{K} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l^\perp)^2, \quad (8)$$

166 where the subspace $\{\mathbf{x}_l^\perp\}$ is obtained by orthogonalising the coverage-weighted predicted linear
167 subspace $\{W^{\frac{1}{2}} \mathbf{x}_l\}$, where $\mathbf{x}_l^T W \mathbf{x}_l = 1$, using the Gram–Schmidt process. This operation ensures
168 that the loss ranges from zero for mutually orthogonalising subspaces to one for identical subspaces
169 and avoids artificially inflating the SS loss due to redundancy in the predicted motions. The order
170 in which the predicted vectors are orthogonalised does not influence the loss, guaranteeing stable
171 training. Appendix A proves this statement.

172 **Independent Subspace (IS) Loss.** We can substitute the orthogonalisation procedure by using an
173 auxiliary loss component for maximising the rank of the predicted subspace. For this purpose, we
174 chose the squared cosine similarity computed between pairs of predicted vectors. The final expression
175 for the *independent subspace* (IS) loss is

$$\text{IS Loss} = \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{x}_k^T W \mathbf{x}_l)^2 - \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l)^2, \quad (9)$$

176 where the predictions $\{\mathbf{x}_l\}$ are normalised prior to the loss computation such that $\mathbf{x}_l^T W \mathbf{x}_l = 1$ and
177 the scaling factor K^2 ensures that the loss ranges between 0 and 1. Appendix A analyses the stability
178 of this formulation.

179 3.4 Architecture

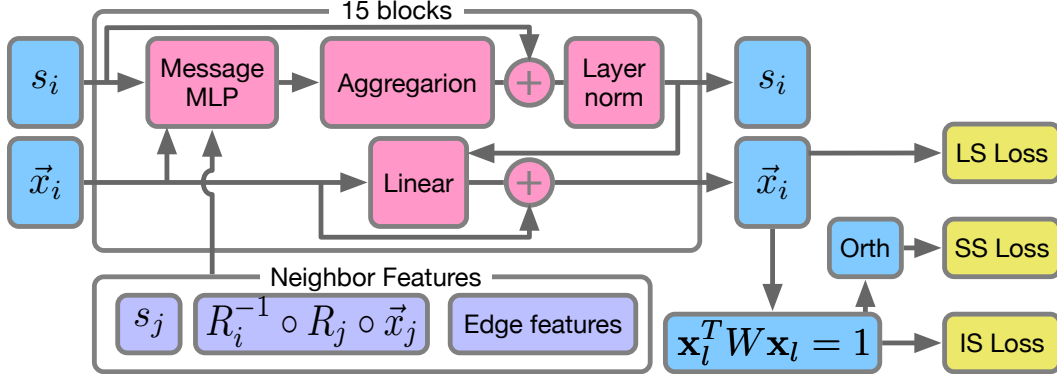


Figure 1: **PETIMOT’s architecture overview.** The model processes both sequence embeddings (s) and motion vectors ($\vec{x}$) through 15 message-passing blocks. Each block updates both representations by aggregating information from neighboring residues. Neighbor features are computed in the reference frame of the central residue i , ensuring SE(3) equivariance. The geometric features encoded in the edges capture the relative spatial relationships between residue pairs. Three types of losses (LS, SS, and IS) are computed, with prior normalization of the predictions for the IS and SS losses, and an additional orthogonalisation of the predictions for the SS loss.

180 **Dual-Track Representation.** PETIMOT processes protein sequences through a message-passing
181 neural network that simultaneously handles residue embeddings and motion vectors in local coordinate
182 frames (Fig. 1). For each residue i , we define and update a node embedding $\mathbf{s}_i \in \mathbb{R}^d$ initialized
183 from protein language model features and a set of K motion vectors $\{\vec{x}_{ik}\}_{k=1}^K \in \mathbb{R}^{3 \times K}$ initialized
184 randomly. The message passing procedure is detailed in Algorithm B.1 of Appendix B.2.

Graph Construction. The protein is represented as a graph where nodes correspond to the residues, and edges capture spatial relationships. For each residue i , we connect to its k nearest neighbors based on $C\alpha$ distances and l randomly selected residues. This hybrid connectivity scheme ensures both local geometric consistency and global information flow, while maintaining sparsity for computational efficiency. Indeed, our model scales *linearly* with the length N of a protein. In our base model we set $k = 5$ and $l = 10$.

Node features. We chose ProstT5 as our default protein language model for initialising node embeddings [57]. This structure-aware pLM offers an excellent balance between model size – including the number of parameters and embedding dimensionality – and performance [17].

Local Reference Frames. Each residue’s backbone atoms (N, CA, C) define a local reference frame through a rigid transformation $T_i \in SE(3)$. For each residue pair (i, j) , we compute their relative transformation $T_{ij} = T_i^{-1} \circ T_j$ from which we extract the rotation $R_{ij} \in SO(3)$ and translation $\vec{t}_{ij} \in \mathbb{R}^3$. Under global rotations and translations of the protein, these relative transformations remain invariant.

Edge Features. Edge features e_{ij} provide an SE(3)-invariant encoding of the protein structure through relative orientations, translational offsets, protein chain distance, and a complete description of peptide plane positioning captured by pairwise backbone atom distances. See Appendix B.3 for more details. The training procedure is detailed in Appendix B.4.

4 Results

Metrics	PETIMOT	AlphaFlow	ESMFlow	NMA
Success Rate (%) $\uparrow$	43.57	31.80	26.82	24.88
Running time $\downarrow$	15.82s	38h 07min	10h 41min	43.59s
Min. LS Error $\downarrow$	0.61 $\pm$ 0.22	0.68 $\pm$ 0.21	0.70 $\pm$ 0.22	0.72 $\pm$ 0.20
Min. Magnitude Error $\downarrow$	0.21 $\pm$ 0.12	0.24 $\pm$ 0.12	0.26 $\pm$ 0.14	0.27 $\pm$ 0.14
OLA LS Error $\downarrow$	0.83 $\pm$ 0.10	0.86 $\pm$ 0.10	0.87 $\pm$ 0.10	0.88 $\pm$ 0.10
OLA Magnitude Error $\downarrow$	0.41 $\pm$ 0.14	0.43 $\pm$ 0.14	0.47 $\pm$ 0.15	0.48 $\pm$ 0.15
Global SS Error $\downarrow$	0.73 $\pm$ 0.14	0.78 $\pm$ 0.14	0.80 $\pm$ 0.14	0.79 $\pm$ 0.14

Table 1: **Success rate and average performance on the test set.** We report performance metrics on a subset of our test set, namely the 824 test proteins that we could process with AlphaFlow (smaller than 450 amino acids). The success rate is defined as the proportion of test proteins with at least one motion predicted at a reasonable accuracy, namely an LS error below 0.6. The other values are averages computed over the test proteins. Min. stands for the best matching pair of predicted and ground-truth vectors. OLA refers to the optimal linear assignment between all predicted and ground-truth vectors. Arrows indicate whether higher ($\uparrow$) or lower ($\downarrow$) metrics values are better. Best results are shown in **bold**. Running times are recorded on a Intel(R) Xeon(R) W-2245 CPU @ 3.90GHz equipped with GeForce RTX 3090. PETIMOT (with 4 directions), AlphaFlow (50 models), and ESMFlow (50 models) were executed on a GPU, while NOLB NMA (with 10 lowest modes) only used CPU.

Training and evaluation. We trained PETIMOT against linear motions extracted from all $\sim 750,000$ protein chains from the PDB (as of June 2023) clustered at 80% sequence identity and coverage. Our full training data comprises 7 335 conformational collections split into 70% for training, 15% for validation and 15% for test. We augmented the training and validation data by computing the motions with respect to 5 reference conformations per collection. As a result, the full training set comprises 25,595 samples. We set the numbers of predicted and ground-truth motions, $K = L = 4$. See Appendix B.1 for more details.

At inference, we consider $w_i = 1, \forall i = 1..N$. We rely on four main evaluation metrics aimed at addressing the following questions:

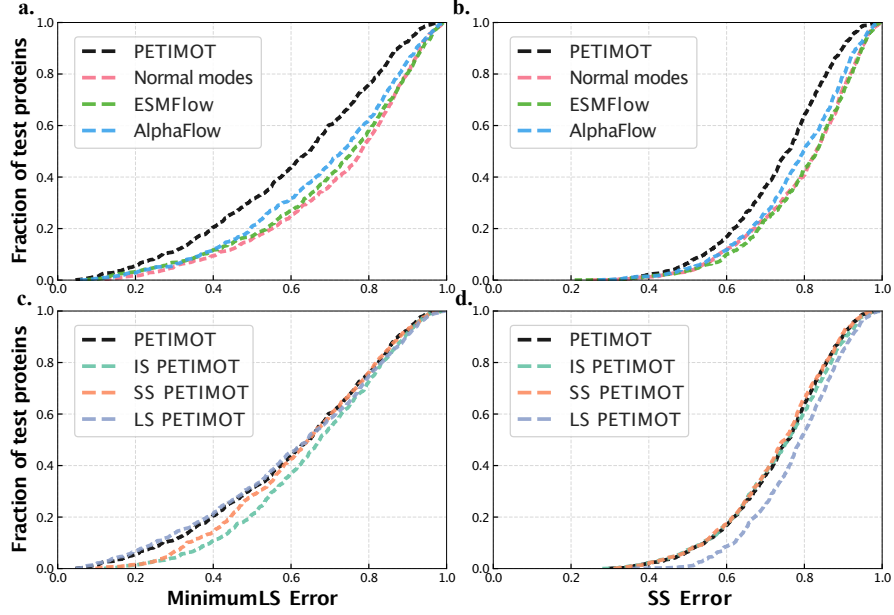


Figure 2: **Cumulative error curves computed on the test proteins.** **a-b.** Comparison between PETIMOT base model and three other methods. **c-d.** Comparison between different losses implemented in PETIMOT. The loss of the base model is LS + SS. **a,c.** Minimum LS error corresponding to the best matching pair of predicted and ground-truth motions. **b,d.** SS error computed between the entire predicted and ground-truth subspaces.

- 213 • Is PETIMOT able to approximate at least one of the main linear motions of a given protein?
- 214 For this, we rely on the minimum LS error over all possible pairs of predicted and ground-
- 215 truth vectors.
- 216 • To what extent does PETIMOT capture the main motion linear subspace of a given protein?
- 217 For this, we use the global SS error.
- 218 • Is PETIMOT able to identify the residues that move the most? Here, we rely on the
- 219 magnitude error, $\frac{1}{N} \sum_{i=1}^N (\|\vec{y}_{ik}\|^2 - \|c_{kl}\vec{x}_{il}\|^2)$.
- 220 • How fast is PETIMOT at inference?

221 **Comparison with other methods.** PETIMOT showed a better capacity to approximate individ-

222 ual motions and to globally capture motion subspaces than the flow matching-based frameworks

223 AlphaFlow and ESMFlow, and also the unsupervised physics-based Normal Mode Analysis (Fig.

224 2a-b). It approximated at least one motion with reasonable accuracy (LS error below 0.6) for 43.57%

225 of the test proteins, while the success rate was only 31.80%, 26.82%, and 24.88% for AlphaFlow,

226 ESMFlow, and the NMA, respectively (Table 1). PETIMOT’s best predicted vector better matched

227 a ground-truth vector than any other methods in 43.57% of the cases (Fig. 3a). PETIMOT was

228 also better at identifying which residues contribute the most to the motions (Table 1). Furthermore,

229 PETIMOT was significantly faster at inference - it took about 16s for the whole test set, followed by

230 NOLB (44s), ESMFlow (11h) and AlphaFlow (38h). See Appendix B.5 for more evaluation details.

231 For all the metrics PETIMOT performs the best, with a particular striking difference in performance

232 for the success rate metrics. However, it maybe not very informative to look at a single value averaged

233 over the whole test set. Thus, we also suggest to analyze more informative plots, *e.g.* those in Fig. 2

234 and Fig. 3a.

235 **Comparison of problem formulations.** Our base model combining the LS and SS losses with

236 equal weights outperforms all three individual losses, LS, SS, and LS (Fig. 2c-d). It strikes an

237 excellent balance between approximating individual motions with high accuracy (Fig. 2c) and

238 globally covering the motion subspaces (Fig. 2d). By comparison, the SS and IS losses tend

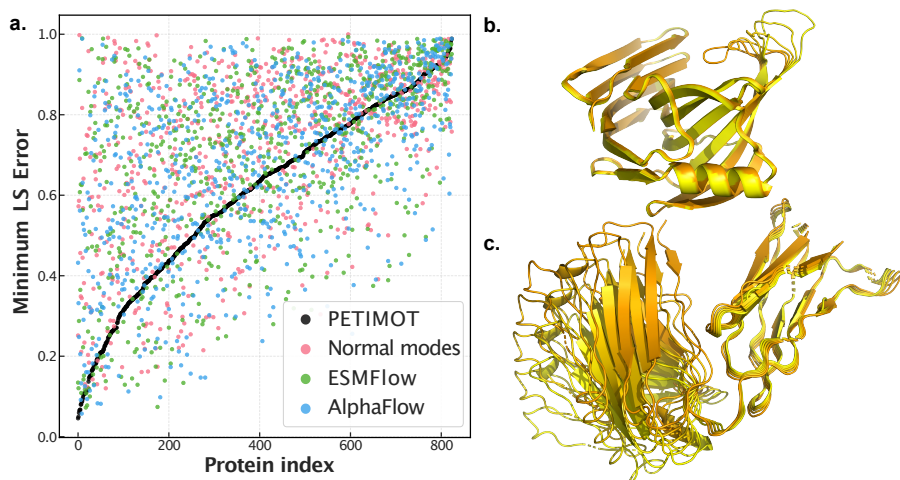


Figure 3: **Individual predictions.** **a.** The per-protein minimum LS errors, computed for the best-matching pairs between predicted and ground-truth vectors, are reported for PETIMOT (black), the NMA (red), AlphaFlow (blue) and ESMFlow (green). The values are in ascending order of the errors computed for PETIMOT, from best to worse. **b-c.** Trajectories generated by deforming a protein structure along PETIMOT best predicted motion. Five trajectory snapshots are shown colored from yellow to orange. **b.** *Bacillus subtilis* xylanase A (PDB id: 3EXU, chain A). **c.** Murine Fab fragment (PDB id: 7SD2, chain A).

239 to underperform on individual motions while the LS loss tends to provide lower coverage of the
240 ground-truth subspaces. See Appendix C for additional results.

241 **Contribution of sequence and structure features.** We performed an ablation study to assess
242 the contribution of sequence and structure information to our architecture. Our results show that
243 ProstT5 slightly outperforms the more recent and larger pLM, ESM-Cambrian 600M [58] (Fig.
244 B.2). Geometrical information about protein structure provides the most significant contribution, as
245 replacing ProstT5 embeddings with random numbers has only a small impact on network performance.
246 Conversely, the network’s performance without structural information strongly depends on the chosen
247 pLM. While the structure-aware embeddings from ProstT5 partially compensate for missing 3D
248 structure information, relying solely on ESM-C embeddings results in poor performance (Fig. B.2).
249 Moreover, connecting each residue to its 15 nearest neighbours (sorted according to $C\alpha$ - $C\alpha$ distances)
250 in the protein graph results in lower performance compared to introducing randomly chosen edges or
251 even fully relying on random connectivity (Fig. B.4).

252 **Generalisation capability.** Because our conformational collections are defined based on 80%
253 sequence identity and coverage thresholds, some test proteins may be close homologs of the training
254 proteins. Yet, the sequence identity and structural similarity (TM-score) of the test proteins with
255 respect to the training set do not determine the quality of PETIMOT predictions (Fig. C.3). PETIMOT
256 provides high-quality predictions for a number of test proteins that do not share any detectable
257 similarity and only weak structural similarity (TM-score below 0.5) to the training set. To further
258 assess PETIMOT’s generalisation capability across protein families, we implemented a more stringent
259 train-validation-test split using a 30% sequence similarity threshold. This process involved three key
260 steps: (1) removing from training and validation sets any proteins sharing >30% sequence similarity
261 with test proteins, (2) redefining these sets through random splitting at the 30% sequence identity level,
262 and (3) reducing redundancy within each set to ensure balanced representation of protein families
263 (detailed in Appendix B.6). When trained under this stringent protocol, PETIMOT achieved a success
264 rate of 40.87% on our non-redundant set of 734 test proteins, consistently outperforming all baseline
265 methods (see Appendix Fig. C.4a-b). This demonstrates our architecture’s robustness, as the more
266 challenging similarity restrictions did not dramatically reduce performance. For direct comparison
267 between our two model versions *PETIMOT-default* (trained with 80% similarity threshold) and
268 *PETIMOT-stringent* (trained with 30% non-redundant threshold) we evaluated performance on 474
269 test proteins that share less than 30% sequence similarity with any protein used in training or

validation of either model. *PETIMOT-stringent* achieved a slightly higher success rate (43.67%) compared to *PETIMOT-default* (41.98%) (see Appendix Fig. C.4c-d). This result emphasizes the benefit of encouraging better generalization to evolutionary distant protein families through enhanced data splitting strategies.

Conformation generation. PETIMOT allows straightforwardly generating conformational ensembles or trajectories by deforming an initial protein 3D structure along one or a combination of predicted motions. We showcase this functionality on two example proteins, the xylanase A from *Bacillus subtilis* and the periplasmic domain of Gliding motility protein GldM from *Capnocytophaga canimorsus* (Fig. 3b-c). We used PETIMOT predictions to generate physically realistic conformations representing either the open-to-closed transition of xylanase A thumb (Fig. 3b) or the flexibility of the heavy chain antibody IgE/Fab anti-profilin Hev b 8 (Fig. 3c). Figures C.5 and C.6 compare predicted motions for these proteins with the ground truth.

Limitations. One current limitation of our approach is the scarcity of functional motions in the training set, raising concerns about its accuracy and completeness. The success rates computed for PETIMOT, although higher than the baseline methods, remain modest, suggesting that some ground-truth motions may lack biological or physical relevance. They may result from experimental artifacts, most often of crystallographic origin. Nevertheless, establishing objective criteria to reliably distinguish artifactual conformations from biologically relevant ones remains infeasible. Our working assumption is that a subset of the conformational manifold in a protein collection always represents functional motions. To address this challenge, we designed our training loss function specifically to evaluate submanifolds by calculating the minimum error between each reference motion and the set of predicted motions, allowing the model to capture conformational diversity while mitigating the impact of potential artefacts.

Another limitation is that we do not account for the different time scales at which different types of protein motions occur. We do not have this temporal information in the PDB training set. The only information we have is the proxy of the amplitudes of the motions, represented by the eigenvalues in the original collections. However, these amplitudes might not be relevant as they strongly depend on the uneven sampling in the PDB and different experimental conditions and temperatures. This motivated us to only reconstruct the shape of the motion manifolds, not their amplitudes or effective timescales.

In addition, our approach is limited to modeling protein motions as linear displacement vectors. While this approximation is sufficient to describe most of the observed conformational heterogeneity, it remains inadequate for modeling highly complex non-linear deformations. Furthermore, deforming proteins structures along linear motion direction may produce unrealistic conformations at large amplitudes. A possible solution yet to be investigated would be the nonlinear extrapolation technique that have been widely used in molecular mechanics [37]. Finally, a direction for future improvement would be to provide per-residue confidence estimates for the predicted motions.

5 Conclusion

In this work, we have proposed a new perspective on the problem of capturing protein continuous conformational heterogeneity. Compared to state-of-the-art methods, our approach goes beyond generating alternative protein conformations by directly inferring compact and continuous representations of protein motions. Our comprehensive analysis of PETIMOT's predictive capabilities demonstrates its performance and utility for understanding how proteins deform to perform their functions. Our work opens ways to future developments in protein motion manifold learning, with exciting potential applications in protein engineering and drug development.

Broader impacts. This paper presents work aimed at advancing the field of computational biology and machine learning, specifically in the context of understanding protein dynamics. While the goal is primarily to enhance scientific understanding and methodological rigor, there are potential societal consequences to consider. These include the facilitation of drug discovery and protein engineering applications, as well as ethical concerns regarding the dual use of protein modeling. We emphasize the importance of using these tools responsibly to maximize benefits while mitigating risks. Beyond these considerations, there are no specific societal implications we feel require urgent attention.

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521 Appendices

522 A Invariance of the proposed losses

523 **Theorem A.1.** *SS Loss is invariant under unitary transformations of X and Y subspaces.*

524 *Proof.* Without loss of generality, let us assume that we apply a unitary transformation $U \in \mathbb{R}^{K \times K}$
 525 to a subspace $X^\perp \in \mathbb{R}^{3N \times K}$, such that the result $X' = X^\perp U$, with $X' \in \mathbb{R}^{3N \times K}$, spans the same
 526 subspace as $X^\perp$, as it is a linear combination of the original basis vectors from $X^\perp$. Then, let us
 527 rewrite the SS loss as

$$\text{SS Loss} = 1 - \frac{1}{K} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l^\perp)^2 = 1 - \frac{1}{K} \|Y^T W^{\frac{1}{2}} X^\perp\|_F^2. \quad (\text{A.1})$$

528 As the Frobenius matrix norm is invariant under orthogonal, or more generally, unitary, transforma-
 529 tions, $\|Y^T W^{\frac{1}{2}} X^\perp U\|_F^2 = \|Y^T W^{\frac{1}{2}} X^\perp\|_F^2$, which completes the proof. $\square$

530 **Corollary A.1.1.** *The SS loss is invariant to the direction permutations in the Gram-Schmidt*
 531 *orthogonalization process.*

532 *Proof.* Let us consider two linear subspaces $X_1^\perp$ and $X_2^\perp$ resulting from the Gram-Schmidt orthogo-
 533 nalization of X , where we arbitrarily choose the order of the orthogonalization vectors. Both $X_1^\perp$
 534 and $X_2^\perp$ will span the same subspace as X , and since both $X_1^\perp$ and $X_2^\perp$ are also orthogonal, one is a
 535 unitary transformation of the other, $X_2^\perp = X_1^\perp U$, which completes the proof. $\square$

536 **Theorem A.2.** *IS Loss is invariant under unitary transformations of X and Y subspaces.*

537 *Proof.* Following the previous proof, without loss of generality, let us assume that we apply an
 538 orthogonal (unitary) transformation $U \in \mathbb{R}^{K \times K}$ to a subspace $X \in \mathbb{R}^{3N \times K}$, such that the result
 539 $X' = XU$, with $X' \in \mathbb{R}^{3N \times K}$, spans the same subspace as X . Then, let us rewrite the IS loss as

$$\text{IS Loss} = \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{x}_k^T W \mathbf{x}_l)^2 - \frac{1}{K^2} \sum_{k=1}^K \sum_{l=1}^K (\mathbf{y}_k^T W^{\frac{1}{2}} \mathbf{x}_l)^2 = \frac{1}{K^2} \|X^T W X\|_F^2 - \frac{1}{K^2} \|Y^T W^{\frac{1}{2}} X\|_F^2. \quad (\text{A.2})$$

540 As the Frobenius matrix norm is invariant under orthogonal transformations, $\|Y^T W^{\frac{1}{2}} XU\|_F^2 =$
 541 $\|Y^T W^{\frac{1}{2}} X\|_F^2$, and $\|U^T X^T W X U\|_F^2 = \|X^T W X\|_F^2$, which completes the proof. $\square$

542 B Methods details

543 B.1 Training data

544 **Conformational collections.** To generate the training data, we utilized DANCE [15] to construct
 545 a non-redundant set of conformational collections representing the entire PDB as of June 2023.
 546 Wherever possible, we enhanced the data quality by replacing raw PDB coordinates with their
 547 updated and optimized counterparts from PDB-REDO [59]. Each conformational collection was
 548 designed to include only closely related homologs, ensuring that any two protein chains within the
 549 same collection shared at least 80% sequence identity and coverage. Collections with insufficient
 550 data points were excluded as we require at least 5 conformations. To simplify the data, we retained
 551 only C α atoms (option -c) and accounted for coordinate uncertainty by applying weights (option
 552 -w).

553 **Handling missing data.** The conformations in a collection may have different lengths reflected
 554 by the introduction of gaps when aligning the amino acid sequences. We fill these gaps with the
 555 coordinates of the conformation used to center the data. In doing so, we avoid introducing biases
 556 through reconstruction of the missing coordinates. Moreover, to explicitly account for data uncertainty,

we assign confidence scores to the residues and include them in the structural alignment step and the eigen decomposition. The confidence score of a position i reflects its coverage in the alignment,

$$w_i = \frac{1}{m} \sum_S \mathbb{1}_{a_i^S \neq "X"}, \quad (\text{B.1})$$

where "X" is the symbol used for gaps and m is the number of conformations. The structural alignment of the j th conformation onto the reference conformation amounts to determining the optimal rotation that minimises the following function [52, 53],

$$E = \frac{1}{\sum_i w_i} \sum_i w_i (r_{ij}^c - r_{i0}^c)^2, \quad (\text{B.2})$$

where r_{ij}^c is the i th centred coordinate of the j th conformation and r_{i0}^c is the i th centred coordinate of the reference conformation. The resulting aligned coordinates are then multiplied by the confidence scores prior to the PCA, as we explain below.

Eigenspaces of positional covariance matrices. The Cartesian coordinates of each conformational ensemble can be stored in a matrix R of dimension $3N \times m$, where N is the number of residues (or positions in the associated multiple sequence alignment) and n is the number of conformations. Each position is represented by a C- α atom. We compute the coverage-weighted (to account for missing data, as explained above) covariance matrix as in Eq. 2. The covariance matrix is a $3N \times 3N$ square matrix, symmetric and real.

We decompose C as $C = VDV^T$, where V is a $3N \times 3N$ matrix with each column defining a sqrt-coverage-weighted eigenvector or a principal component that we interpret as a *linear motion*. D is a diagonal matrix containing the eigenvalues. Specifically, the k th principal component was expressed as a set of 3D (sqrt-coverage-weighted) displacement vectors $\tilde{x}_{ik}^{GT}$, $i = 1, 2, \dots, L$ for the L C α atoms of the protein residues. To enable cross-protein comparisons, the vectors were normalized such that $\sum_i \tilde{x}_{ik}^{GT} \cdot \tilde{x}_{ik}^{GT} = L$. The sum of the eigenvalues $\sum_{k=1}^{3m} \lambda_k$ amounts to the total positional variance of the ensemble (measured in $\text{\AA}^2$) and each eigenvalue reflects the amount of variance explained by the associated eigenvector.

Data augmentation. The reference conformation used to align and center the 3D coordinates corresponds to the protein chain with the most representative amino acid sequence. To increase data diversity, four additional reference conformations were defined for each collection. At each iteration, the new reference conformation was selected as the one with the highest RMSD relative to the previous reference. This iterative strategy maximizes the variability of the extracted motions by emphasizing the impact of changing the reference.

B.2 Message passing

The node embeddings and predicted motion vectors are updated iteratively according to the following algorithm.

Algorithm B.1 PETIMOT Message Passing Block

```
1: function MESSAGEPASSING( $\{\mathbf{s}_i\}, \{\vec{x}_i\}, \{\mathcal{N}eigh(i)\}, \{R_{ij}, e_{ij}\}$ ):  
2: #  $\{\mathbf{s}_i\}_{i=1}^N$  ▷ Node embeddings  
3: #  $\{\vec{x}_i\}_{i=1}^N$  ▷ Motion vectors in local frames  
4: #  $\{\mathcal{N}eigh(i)\}_{i=1}^N$  ▷ Node neighborhoods  
5: #  $\{R_{ij}, e_{ij}\}$  ▷ Relative geometric features  
6:   for  $i = 1$  to  $N$  do  
7:     for  $j \in \mathcal{N}eigh(i)$  do  
8:        $\vec{x}_j^i \leftarrow R_{ij}\vec{x}_j$  ▷ Project motion in frame  $i$   
9:        $m_{ij} \leftarrow \text{MessageMLP}(\mathbf{s}_i, \mathbf{s}_j, \vec{x}_i, \vec{x}_j^i, e_{ij})$   
10:    end for  
11:     $m_i \leftarrow \text{Mean}_j(m_{ij})$  ▷ Aggregate messages  
12:     $\mathbf{s}_i \leftarrow \mathbf{s}_i + \text{LayerNorm}(m_i)$  ▷ Update embedding  
13:     $\vec{x}_i \leftarrow \vec{x}_i + \text{Linear}([\mathbf{s}_i, \vec{x}_i])$  ▷ Update motion  
14:  end for  
15:  return  $\{\mathbf{s}_i\}_{i=1}^N, \{\vec{x}_i\}_{i=1}^N$   
16: end function
```

B.3 SE(3)-equivariant features

We represent protein structures as attributed graphs. The node embeddings are computed with the pre-trained protein language model ProST5 [57]. It is a fine-tuned version of the sequence-only model T5 that translates amino acid sequences into sequences of discrete structural states and reciprocally.

The edge embeddings are computed using SE(3)-invariant features derived from the input backbone, similarly to prior works [31, 48, 46]. Specifically, the features associated with the edge e_{ij} from node (atom) i to node (atom) j are:

- **Quaternion representation:** A 4-dimensional quaternion encoding the relative rotation R_{ij} between the local reference frames of residues i and j .
- **Relative translation:** A 3-dimensional vector representing the translation $\vec{t}_{ij}$ between the local reference frames.
- **Chain separation:** The sequence separation between residues i and j , encoded as $\log(|i - j| + 1)$.
- **Spatial separation:** The logarithm of the Euclidean distance between residues i and j , computed as $\log(\|\vec{t}_{ij}\| + \epsilon)$, where $\epsilon = 10^{-8}$.
- **Backbone atoms distances:** Distances between all backbone atoms (N, C α , C, O) at residues i and j , encoded through a radial basis expansion. For each pairwise distance d_{ab} , we compute:

$$f_k(d_{ab}) = \exp\left(-\frac{(d_{ab} - \mu_k)^2}{2\sigma^2}\right), \quad (\text{B.3})$$

where $\{\mu_k\}_{k=1}^{20}$ are centers spaced linearly in $[0, 20]$ Å and $\sigma = 1$ Å. This creates a $16 \times 20 = 320$ dimensional feature vector, as we have 16 pairwise distances (4×4 atoms) each expanded in 20 basis functions.

B.4 Training procedure

We randomly split the 7,335 conformational collections defined with DANCE into training, validation, and test sets with a ratio 70:15:15. The data augmentation procedure resulted in $5,119 \times 5 = 25,595$ training samples and $1,099 \times 5 = 5,495$ validation samples.

The model was optimized using AdamW [60] with a learning rate of 5e-4 and weight decay of 0.01. We employed gradient clipping with a maximum norm of 10.0 and mixed precision training with PyTorch’s Automatic Mixed Precision. The learning rate was adjusted using torch’s ReduceLROnPlateau scheduler, which monitored the validation loss, reducing the learning rate by a factor of 0.2 after 10 epochs without improvement. Training was performed with a batch size of 32 for both

618 training and validation sets. We implemented early stopping with a patience of 50 epochs, monitoring
619 the validation loss. The model achieving the best validation performance was selected for final
620 evaluation. We trained the model on a single NVIDIA A100-SXM4-80GB GPU. One epoch took
621 about 9 minutes of real time.

622 B.5 Evaluation procedures

623 We report all evaluation metrics on 824 test proteins out of a total of 1 117. The 293 protein
624 we excluded were too long (>450 amino acids) to be handled by AlphaFlow and ESMFlow in a
625 reasonable amount of time using our computing resources.

626 **Comparison with AlphaFlow and ESMFlow.** We compared our approach with the flow-matching
627 based frameworks AlphaFlow and ESMFlow for generating conformational ensembles. For this,
628 we downloaded the distilled "PDB" models from <https://github.com/bjing2016/alphafold>.
629 We executed AlphaFlow using the following command,

```
630 python predict.py --noisy_first --no_diffusion --mode alphafold  
631 --input_csv seqs.csv --msa_dir msa_dir/  
632 --weights alphafold_pdb_distilled_202402.pt --samples 50  
633 --outpdb output_pdb/
```

634 AlphaFlow relies on OpenFold [61] to retrieve the input multiple sequence alignment (MSA). ESM-
635 Flow was launched using the same command with an additional `--mode esmfold` flag and its
636 corresponding weights. We used AlphaFlow and ESMFlow to generate 50 conformations for each
637 test protein and then we treated each ensemble as a conformational collection. We then aligned all
638 members of the created collections to the reference conformations of the ground-truth collections.
639 We used the identity coverage weights here. Finally, from the aligned collections, we extracted the
640 principal linear motions. We shall additionally mention that we did not filter or adapt our test set to
641 the AlphaFlow and ESMFlow methods. In other words, there can be certain data leakage between
642 AlphaFlow/ESMFlow train data and our test examples.

643 **Comparison with the Normal Mode Analysis.** We also compared our approach with the physics-
644 based unsupervised Normal Mode Analysis (NMA) method [38]. The NMA takes as input a protein
645 3D structure and builds an elastic network model where the nodes represent the atoms and the
646 edges represent springs linking atoms located close to each other in 3D space. The four lowest
647 normal modes are obtained by diagonalizing the mass-weighted Hessian matrix of the potential
648 energy of this network. We used the highly efficient NOLB method, version 1.9, downloaded from
649 <https://team.inria.fr/nano-d/software/nolb-normal-modes/> [37] to extract the first K
650 normal modes from the test protein 3D conformations. Specifically, we used the following command

```
651 NOLB INPUT.pdb -c 10 -x -n 4 --linear -s 0 --format 1 --hetatm
```

652 We retained only the $C\alpha$ atoms and defined the edges in the elastic network using a distance cutoff of
653 10 Å.

654 B.6 Generalisability assessment across protein families

655 Protein folds are not evenly represented in the PDB, with some folds being significantly more
656 abundant than others. For our default training procedure, we chose not to correct for this bias because
657 the same fold may exhibit different motions in different collections. This represents a fundamental
658 difference from structure prediction tasks, where distant homologs often exhibit nearly identical 3D
659 structures. In contrast, for protein motion prediction, even proteins with the same fold can display
660 dramatically different conformational changes depending on their specific biological context, binding
661 partners, or experimental conditions. Furthermore, the diversity of motions within a single fold family
662 provides valuable information for our model to learn the relationship between sequence, structure,
663 and dynamic behavior.

664 Nevertheless, to rigorously assess PETIMOT’s capability to generalise across protein families, we
665 re-trained and re-evaluated the model using a more stringent and non-redundant training-validation-
666 test splitting scheme. Specifically, we considered clusters of protein chains defined at 30% sequence

identity and 80% sequence coverage using MMseqs2 [62]. These clusters define distant protein families and we refer to them as *clus-30* in the following. We kept the 824 test proteins used for comparison with other methods as is and we removed all collections belonging to the same *clus-30* clusters as these proteins from the training and validation sets. Then, we re-defined a training-validation random split at the level of the *clus-30* clusters with a 9:1 ratio. This operation ensures that any pair of training-validation, training-test or validation-test collections do not share more than 30% sequence identity. Finally, for each training or validation *clus-30* cluster, we randomly drew 5 samples. This step ensures that each protein family is evenly represented in the training and validation sets. We also redundancy reduced the test set of 824 proteins (test-824), keeping only one protein for each *clus-30* cluster, which led to 734 proteins (test-734).

To be able to compare the model’s default version, *PETIMOT-default*, and re-trained version, *PETIMOT-stringent*, we computed their performance metrics on a subset of 474 test proteins from test-734. These proteins do not fall in any of the *clus-30* clusters used for training or validation any of the two versions.

B.7 Ablation Studies

To understand the impact of different components on the performance of our model, we carried out ablation studies. We list them below.

Model architecture variations.

- Network depth: We experimented with different numbers of message-passing layers (5 and 10 layers compared to our default value of 15 layers).
- Layer sharing: We tested a variant where all message-passing layers share the same parameters, as opposed to our default where each layer has unique parameters.
- Reduced internal embedding dimension: We tested a model with a smaller internal embedding dimension of 128 instead of the default 256.

Figure B.1 shows the evaluation of these modifications. A shallow 5-layers network underperforms on all evaluation metrics. The difference between other variants is not very significant.

Structure and sequence information ablation.

- Structure ablation: We removed all structural information from the model to assess the importance of geometric features and the performance with the PLM embeddings only. We did it by removing the edge attributes of the input of the message passing MLP.
- Sequence ablation: We ablated sequence information by replacing protein language model embeddings with random embeddings, testing them both with and without structural information.
- Embedding variants: We evaluated a different protein language model (ESMC-600M), both with and without structural tokens.

The evaluation results are shown in Fig. B.2. The results demonstrate that while both ProST5 and ESM-Cambrian 600M perform similarly when combined with structural information, removing structural features leads to markedly different outcomes. ProST5 embeddings partially compensate for the missing structural information, likely due to their structure-aware training, while relying solely on ESM-C embeddings results in poor performance.

Problem formulation ablation. We analyzed different combinations of our loss terms (compared to our default balanced weights of LS + SS):

- Least Square loss (LS): Using only the LS loss (weight 1.0).
- Squared Sinus loss (SS): Using only the SS loss (weight 1.0).
- Independent Subspaces (IS): Using only the IS loss (weight 1.0).

Figure B.3 compares three individual losses with the default option. The IS problem formulation underperforms on all the metrics. The default LS + SS formulation performs slightly better than those with individual loss components.

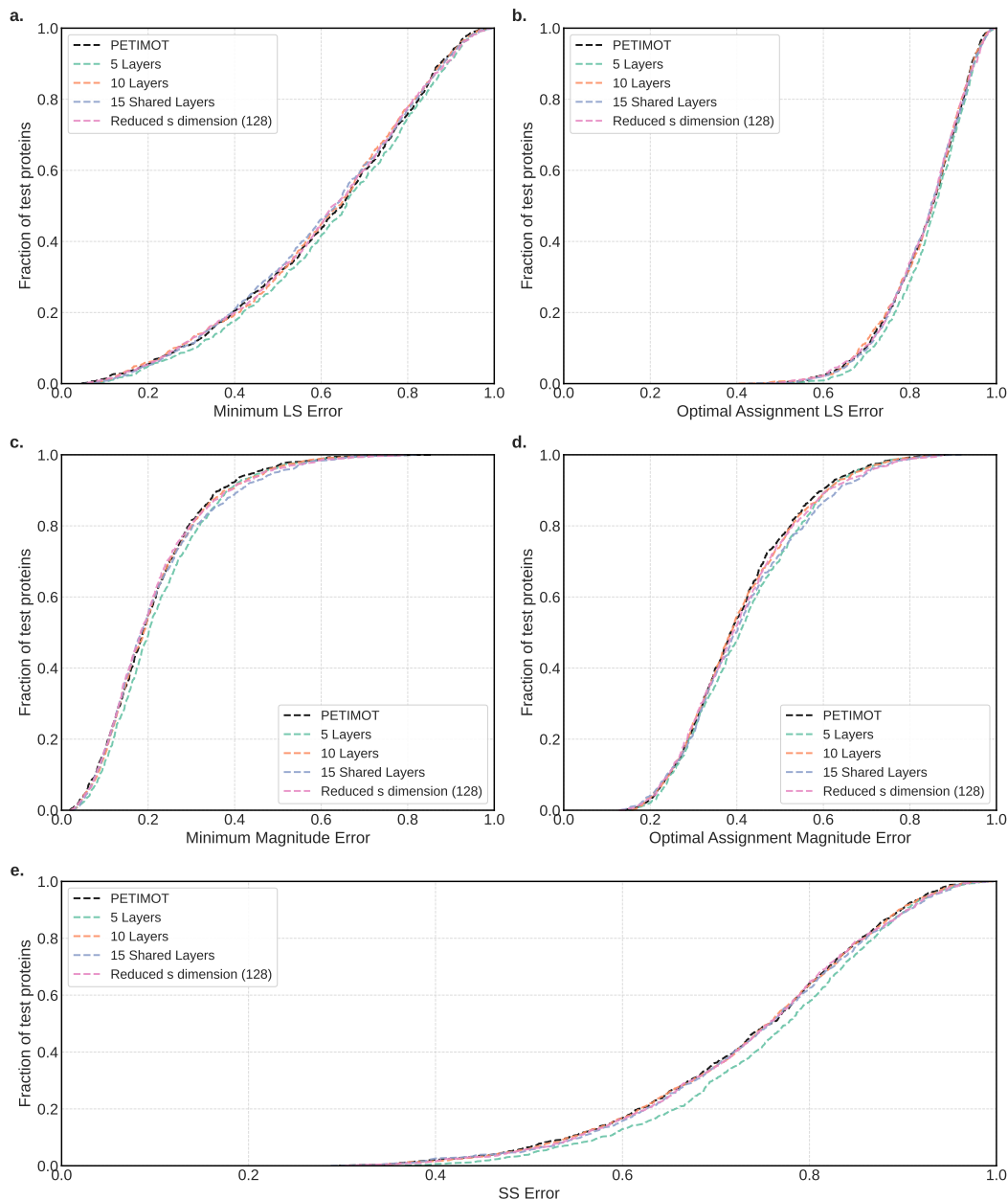


Figure B.1: Network depth ablation. We report cumulative curves for LS error (a-b), magnitude error (c-d), and SS error (e). For each protein, we computed the error either for the best-matching pair of predicted and ground-truth vectors (a,c) or for the best combination of four pairs of predicted and ground-truth vectors (b,d). We vary the number of layers in the network and the embedding dimension.

Graph connectivity ablation. We investigated different approaches to constructing the protein graph:

- Nearest neighbor-only: Using 15 nearest neighbors (sorted according to the corresponding $C\alpha$ - $C\alpha$ distances) without random edges.
- Random connections-only: Using 15 random edges without nearest neighbors. This set is updated between every layer at each epoch.

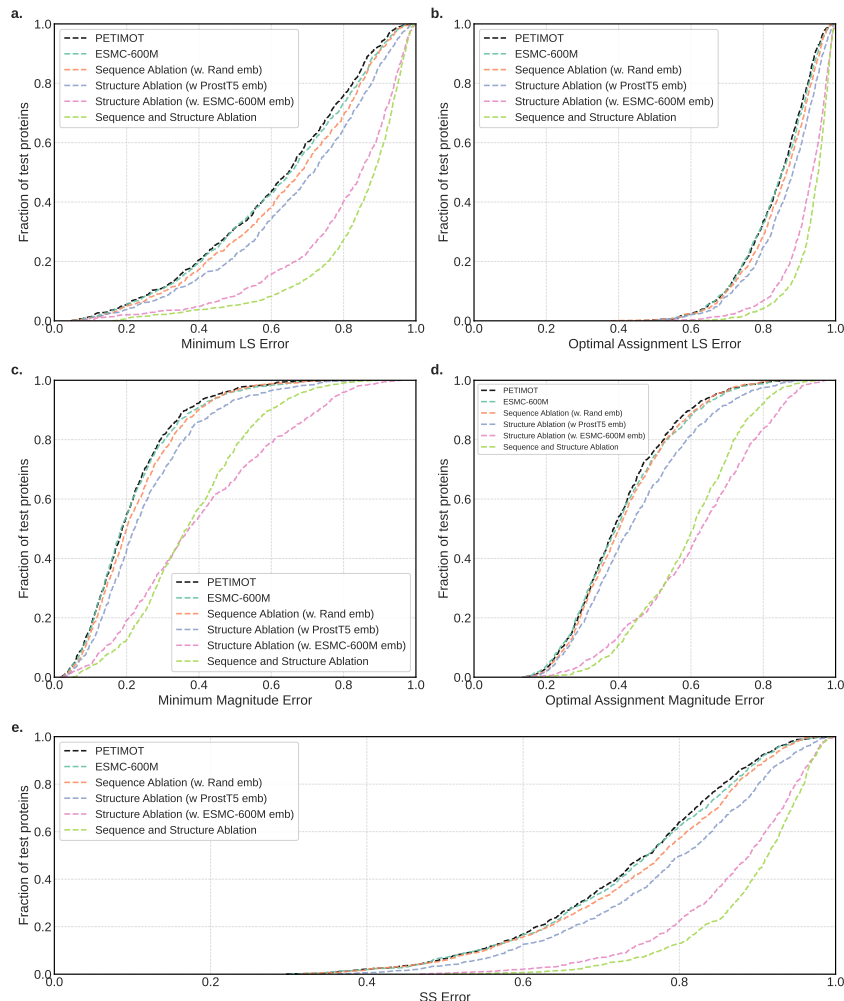


Figure B.2: **Structure and sequence information ablation study.** We report cumulative curves for LS error (a-b), magnitude error (c-d), and SS error (e). For each protein, we computed the LS and magnitude errors either for the best-matching pair of predicted and ground-truth vectors (a,c) or for the best combination of four pairs of predicted and ground-truth vectors (b,d).

- Static connectivity: Using a fixed set of random neighbors between the layers. This set is updated at each epoch.

Figure B.4 shows the ablation results. We can see that the nearest neighbor-only setup underperforms on all the metrics. Among other options, the random connectivity-only option gets lower results at higher metrics values. The default option performs on par with the static connectivity, showing slightly better results on the optimal assignment magnitude error metrics.

C Additional results

Figure C.1 evaluates PETIMOT against NMA, ESMFlow and AlphaFlow approaches using additional metrics. These include the minimum magnitude error, the optimal assignment magnitude error, and the optimal assignment LS error. On all the metrics we see that PETIMOT outperforms the three other tested approaches.

We also experimented with a different number of predicted components. For these experiments, we trained additional models with the LS loss only, which are listed below:

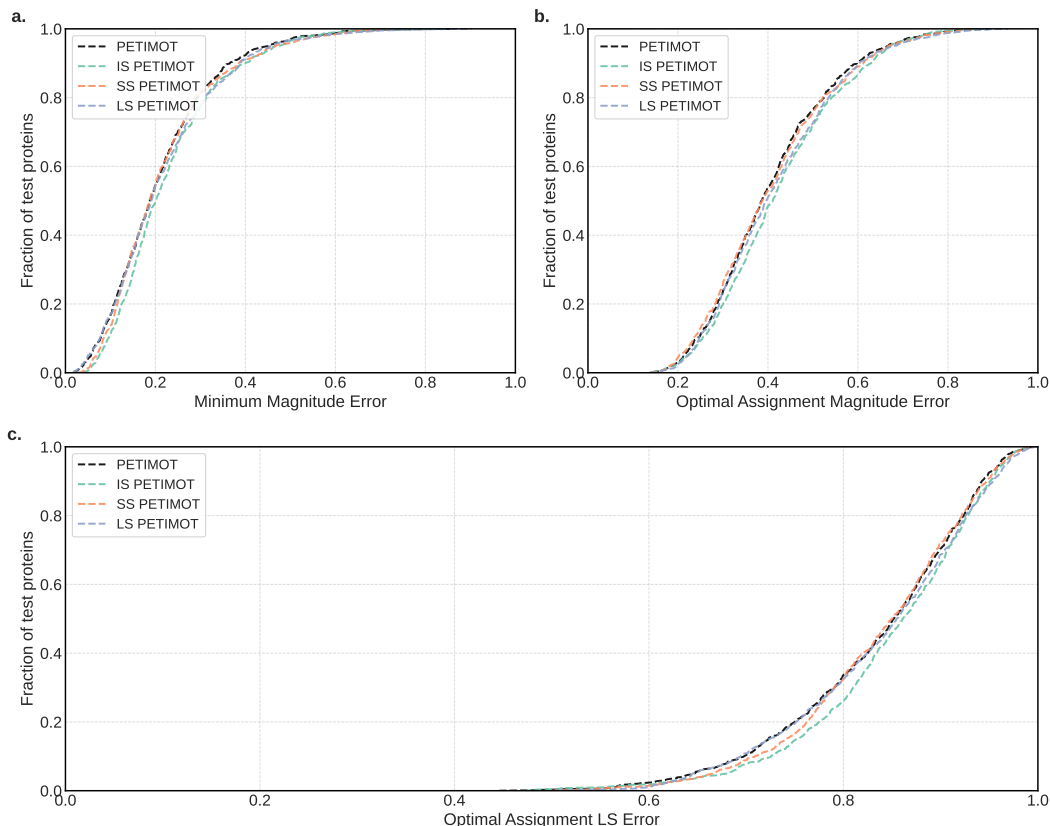


Figure B.3: **Performance comparison of different problem formulations.** We report cumulative curves for magnitude error (a,b) and LS error (c). For each protein, we computed the error either for the best-matching pair of predicted and ground-truth vectors (a) or for the best combination of four pairs of predicted and ground-truth vectors (b,c).

- 734 • Single component prediction (1 mode).
- 735 • Reduced component prediction (2 modes).
- 736 • Extended component prediction (8 modes).

737 We compare these options with our default setting of 4 components. Figure C.2 shows the results.
 738 Increasing the number of predicted components from 1 to 8 improves the minimum LS errors, as
 739 having more predicted vectors naturally increases the chance of matching at least one ground-truth
 740 motion well. However, when evaluating the optimal linear assignment metrics, which measures
 741 overall subspace alignment, models with 1 or 2 components have an artificial advantage since they
 742 face fewer matching constraints. The 8-components model similarly benefits from having more
 743 candidate vectors to match against the 4 ground-truth components.

744 Figure C.3 compares the accuracy of the predicted test proteins (minimum LS loss) with the structural
 745 (TM-score) and sequence (sequence identity) distances to the training set. We do not see a clear
 746 correlation between the prediction accuracy and the similarity to the training examples. Please also
 747 see Fig. 3b-c for comparison.

748 Figure C.4 further assesses PETIMOT’s generalisation capability across protein families. To do
 749 so, we implemented a more stringent train-validation-test split using a 30% sequence similarity
 750 threshold (see main text and Appendix B.6). When trained under this stringent protocol, PETIMOT
 751 still substantially outperforms all baseline methods according to minimum L and SS metrics (Fig.
 752 C.4a-b). Moreover, we observe a slight generalisation improvement of *PETIMOT-stringent* over

753 *PETIMOT-default* on a test set of 474 proteins evolutionary distant from any protein used for training
754 or validation of any of the two model versions (Fig. C.4c-d).

755 Figures C.5 and C.6 show predicted (blue arrows) and ground-truth (red arrows) motion vectors for
756 the xylanase A from *Bacillus subtilis* and the periplasmic domain of Gliding motility protein GldM
757 from *Capnocytophaga canimorsus*, respectively.

758 D Licenses for used resources

759 In this work, we utilize several existing resources. The protein structures were obtained from
760 the Protein Data Bank (PDB, <https://www.rcsb.org/>, version accessed on June 2023) which is dis-
761 tributed under the CC0 1.0 Universal Public Domain Dedication license (CC0 1.0). We com-
762 plemented PDB data with data from PDB-redo (accessed June 2023) developed by Joosten *et*
763 *al.* [59], available at <https://pdb-redo.eu> under the licence specified at <https://pdb-redo.eu/license>. For protein language modeling, we employed ProstT5 developed by Heinzinger
764 *et al.* [57], available under the MIT license at <https://huggingface.co/Rostlab/ProstT5>,
765 and ESM-Cambrian 600M (version esmc-600m-2024-12) developed by the EvolutionaryScale
766 Team [58], available under the Cambrian Non-Commercial License at [https://huggingface.co/](https://huggingface.co/EvolutionaryScale/esmc-600m-2024-12)
767 [EvolutionaryScale/esmc-600m-2024-12](https://huggingface.co/EvolutionaryScale/esmc-600m-2024-12). Additional resources include the DANCE method
768 (version of Oct 8, 2024) developed by Lombard *et al.*, available under the MIT license at
769 <https://github.com/PhyloSofS-Team/DANCE>. As baselines, we ran the NOLB method (ver-
770 sion 1.9) developed by Hoffmann *et al.* [37] and available at [https://team.inria.fr/nano-d/](https://team.inria.fr/nano-d/software/nolb-normal-modes/)
771 [software/nolb-normal-modes/](https://team.inria.fr/nano-d/software/nolb-normal-modes/), and the AlphaFlow (version AlphaFlow-PDB distilled) and
772 ESMFlow (version ESMFlow-PDB distilled) models developed by Jing *et al.* [28] and available at
773 <https://github.com/bjing2016/alphafLOW>. We used TM-align (version 20220412) developed
774 by Zhang and Skolnick [63] and available at <https://zhanggroup.org/TM-align/> to perform
775 all-to-all pairwise structural alignments between train and test protein conformations and compute
776 TM-scores.
777

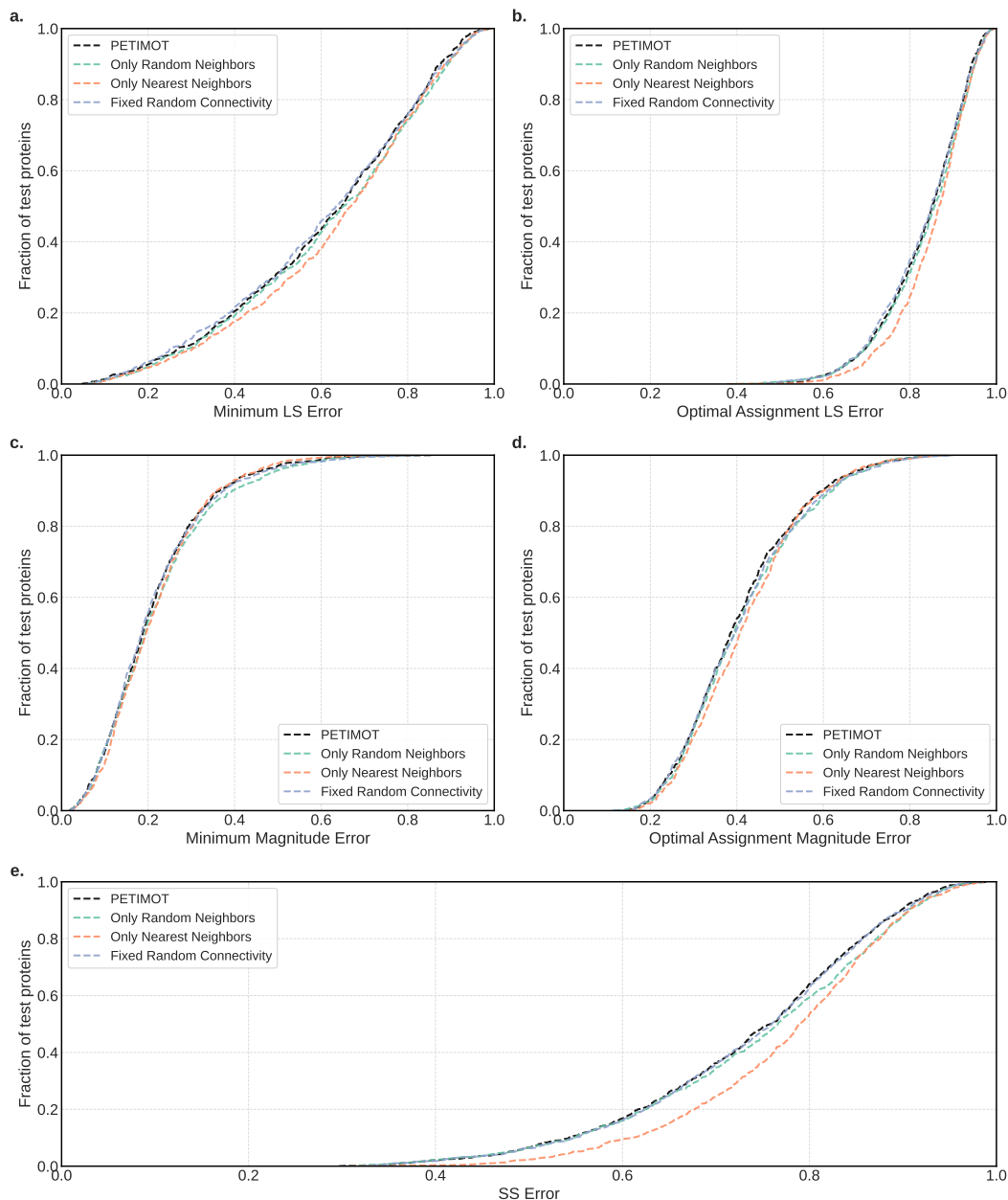


Figure B.4: Graph connectivity ablation. We report cumulative curves for LS error (a-b), magnitude error (c-d), and SS error (e). For each protein, we computed the error either for the best-matching pair of predicted and ground-truth vectors (a,c) or for the best combination of four pairs of predicted and ground-truth vectors (b,d). Only Random Neighbors: each residue (node) is connected to 15 randomly chosen residues and the connectivity changes after each layer. Only Nearest Neighbors: each residue (node) is connected to its 15 nearest neighbors in the input 3D structure. Fixed Random Connectivity: each residue (node) is connected to 15 residues randomly chosen at the beginning.

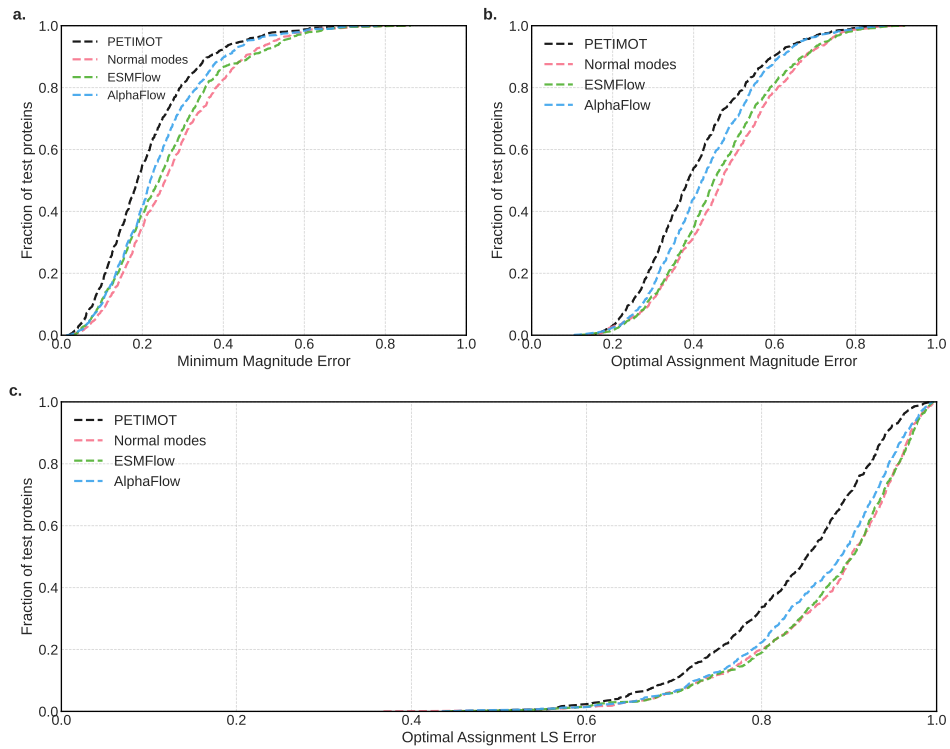


Figure C.1: Performance comparison with other methods on the test proteins. We report cumulative curves for magnitude error (a,b) and LS error (c). For each protein, we computed the error either for the best-matching pair of predicted and ground-truth vectors (a) or for the best combination of four pairs of predicted and ground-truth vectors (b,c).

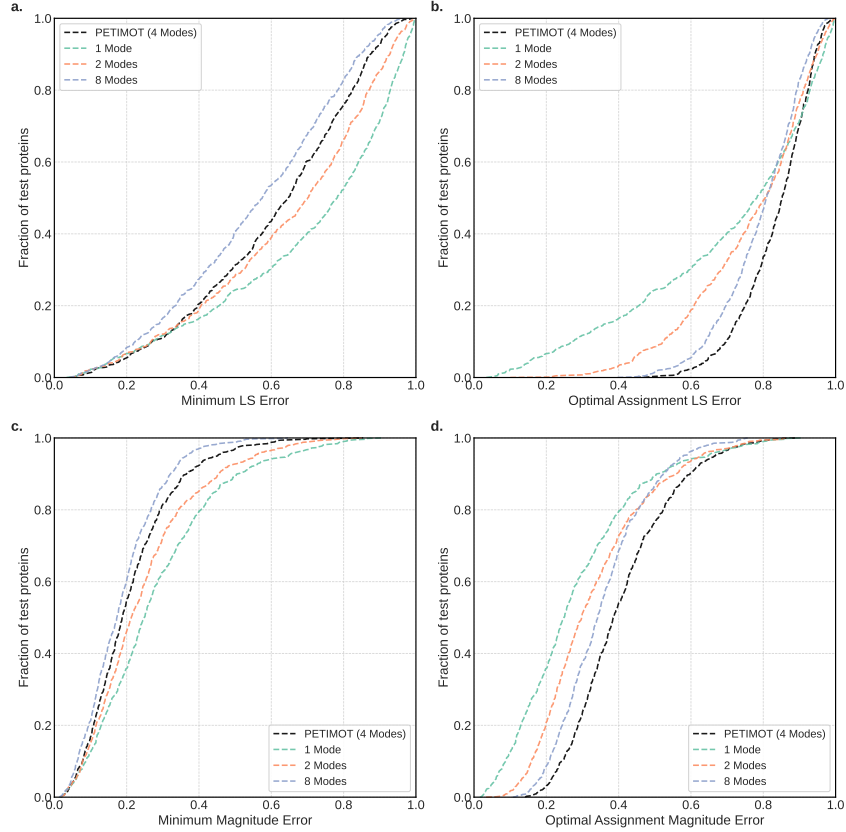


Figure C.2: Impact of the number of predicted components. We report cumulative curves for LS error (a-b) and magnitude error (c-d). For each protein, we computed the error either for the best-matching pair of predicted and ground-truth vectors (a,c) or for the best combination of all pairs of predicted and ground-truth vectors using optimal linear assignment (b,d). We compare models trained to predict different numbers of components (modes): 1, 2, 4, or 8, using only the LS loss.

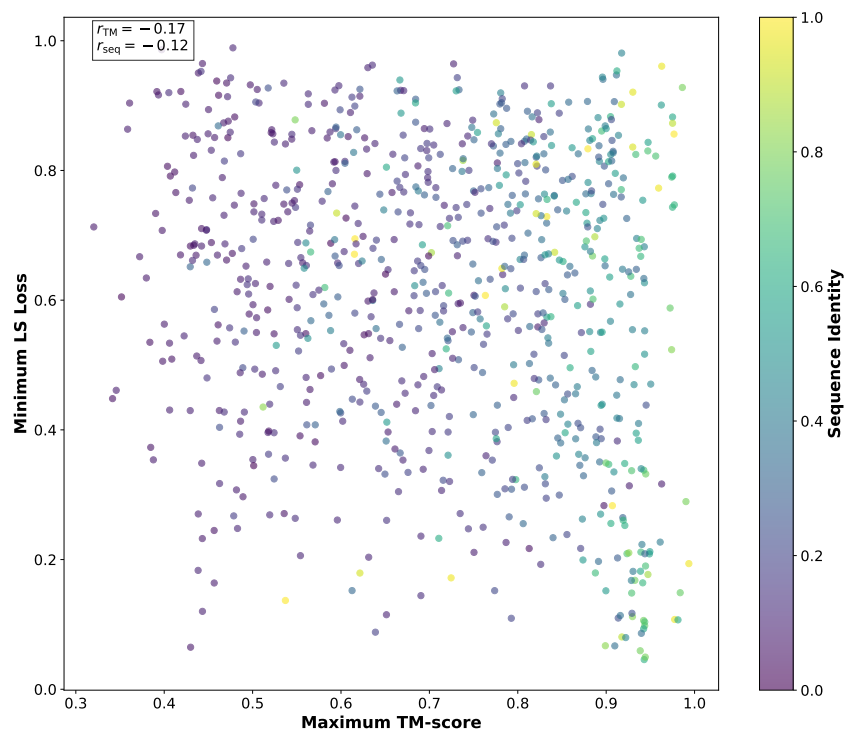


Figure C.3: **Relationship between PETIMOT's prediction accuracy and structural/sequence similarity with the training set.** The minimum LS error is plotted against the maximum TM-score between each test protein and any protein in the training set. Points are colored by the maximum sequence identity to the training samples.

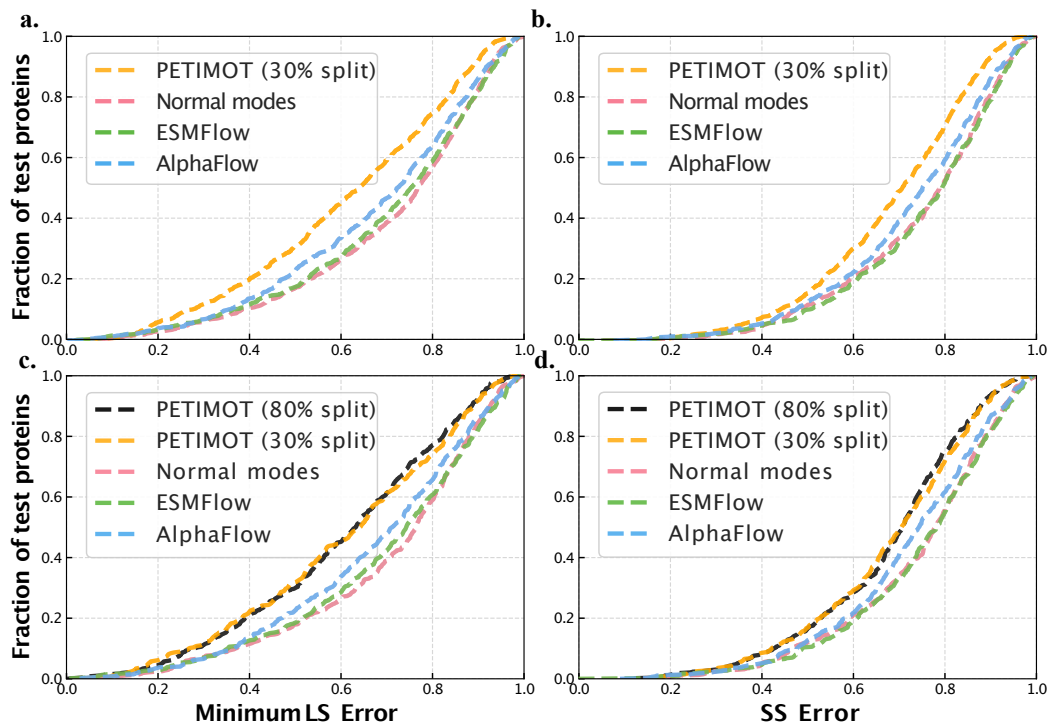


Figure C.4: **Cumulative error curves computed on the test proteins.** **a-b.** Comparison between *PETIMOT-stringent* model and three other methods on a non-redundant set of 734 test proteins. *PETIMOT-stringent* was trained on a stringent and non-redundant training-validation-test split defined using a 30% sequence identity threshold. **c-d.** Comparison between *PETIMOT-default*, *PETIMOT-stringent* and three other methods on 474 test proteins that share less than 30% sequence similarity with any protein used in training or validation of either model. **a,c.** Minimum LS error corresponding to the best matching pair of predicted and ground-truth motions. **b,d.** SS error computed between the entire predicted and ground-truth subspaces.

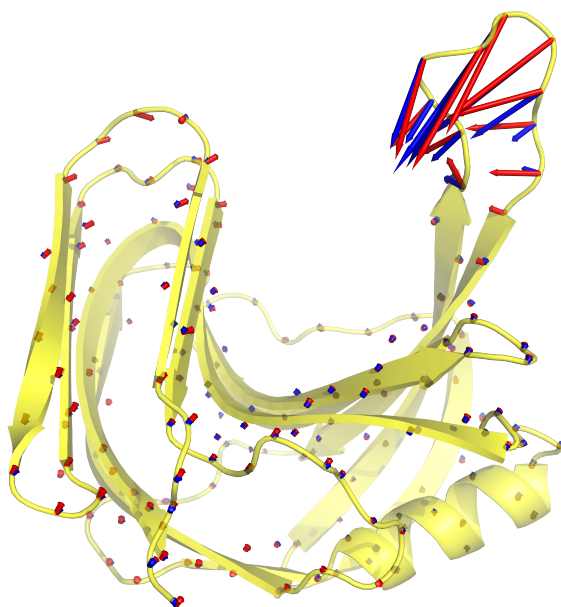


Figure C.5: **Visualization of predicted (blue arrows) and ground-truth (red arrows) motion vectors for PDB structure 3EXU (chain A), with LS error of 0.20.** The predicted deformation was used to generate the interpolated conformations shown in Fig. 3b.

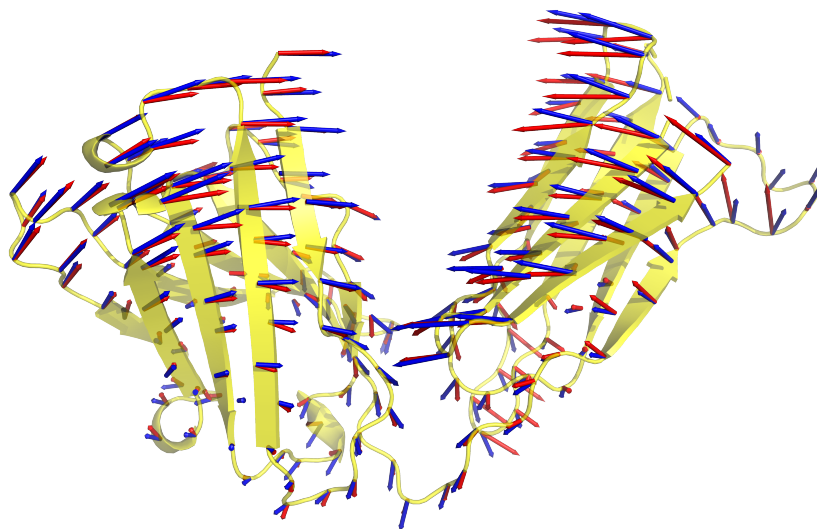


Figure C.6: **Visualization of predicted (blue arrows) and ground-truth (red arrows) motion vectors for PDB structure 7SD2, with LS error of 0.18.** The predicted deformation was used to generate the interpolated conformations shown in Fig. 3c.

NeurIPS Paper Checklist

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Answer: [Yes]

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